

The world's oceans

THERE ARE FIVE OCEANS—THE PACIFIC, ATLANTIC, INDIAN, ARCTIC, AND SOUTHERN. The first four fill natural basins in the Earth's crust. The Southern Ocean is technically part of the southern Pacific, Atlantic, and Indian Oceans but was officially delimited from them in 2000 south of 60 degrees latitude, by the International Hydrographic Organization. This coincides with the limits of the Antarctic Treaty.



PACIFIC OCEAN

The Pacific Ocean is the world's largest ocean, covering about 28% of the Earth's surface. The ocean has between 20,000 and 30,000 islands, and is surrounded by a "Ring of Fire," where the Earth's tectonic plates are sliding into ocean trenches, causing volcanic activity and frequent earthquakes.

AREA: 58,957,258 sq miles (152,617,160 sq km)

Includes: Bali Sea, Bering Sea, Bering Strait, Coral Sea, East China Sea, Florest Sea, Gulf of Alaska, Gulf of Tonkin, Java Sea, Philippine Sea, Savu Sea, Sea of Japan, Sea of Okhotsk, South China Sea, Tasman Sea, Timor Sea

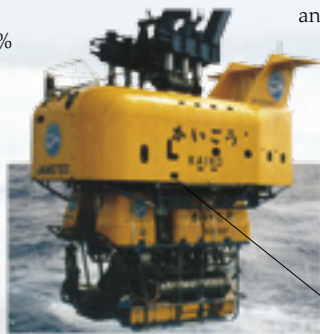
AVERAGE DEPTH: 13,874 ft (4,229 m)

DEEPEST POINT: 36,201 ft (11,034 m)

Challenger Deep in the Mariana Trench

COASTLINE: 84,299 miles (135,663 km)

CLIMATE: Strong currents and trade winds constantly blow across the Pacific's waters, affecting its climate and weather



Kaiko submersible

and often causing violent tropical storms. A cold current usually flows from the western coast of South America. However, every few years a warm current (El Niño) flows east toward Peru, causing worldwide weather changes.

Unmanned sub reached bottom of the Mariana Trench in 1995

NATURAL RESOURCES:

Fish stocks, oil, and gas fields, sand, and gravel aggregates.

Coral atoll reef in the southwest Pacific

ENVIRONMENTAL ISSUES: Nearly half of the world's shipping routes cross the Pacific, including huge supertankers, giant bulk carriers and container ships. As a result, the ocean suffers from oil pollution, especially in the Philippine Sea and the South China Sea, which threatens marine life and seabirds. Some of the Pacific's endangered marine creatures include dugongs, sea otters, sea lions, seals, turtles, and whales.



ATLANTIC OCEAN

The Atlantic Ocean is the world's second-largest ocean, covering about one-fifth of the Earth's surface. An underwater mountain chain called the Mid-Atlantic Ridge runs down its center.

Panama Canal links the Atlantic and Pacific



AREA: 31,563,463 sq miles (81,527,400 sq km)

Includes: Baltic Sea, Black Sea, Caribbean Sea, Davis Strait, Denmark Strait, Gulf of Guinea, Gulf of Mexico, Labrador Sea, Mediterranean Sea, North Sea, Norwegian Sea, Sargasso Sea, Scotia Sea

AVERAGE DEPTH: 12,391 ft (3,777 m)

DEEPEST POINT: 28,232 ft (8,605 m)

Milwaukee Deep in the Puerto Rico Trench

COASTLINES: 69,512 miles (111,866 km)

CLIMATE: The Atlantic's northerly waters are usually covered with sea ice in the northern winter, and huge icebergs drifting southwards are sometimes a problem to shipping from February to August. The Gulf Stream, a warm water current, flows from the Gulf of Mexico, north and then east, feeding the North Atlantic Drift, which raises the temperatures of northern Europe and keeps many northern ports ice-free during the winter.

NATURAL RESOURCES:

Fish, oil, and gas fields, sand, and gravel aggregates.

ENVIRONMENTAL ISSUES:

Some Atlantic waters are polluted from industrial waste, sewage, and oil. Fish stocks have run low because of overfishing, especially through trawling for bottom-dwelling fish such as cod.



Atlantic fishing trawler